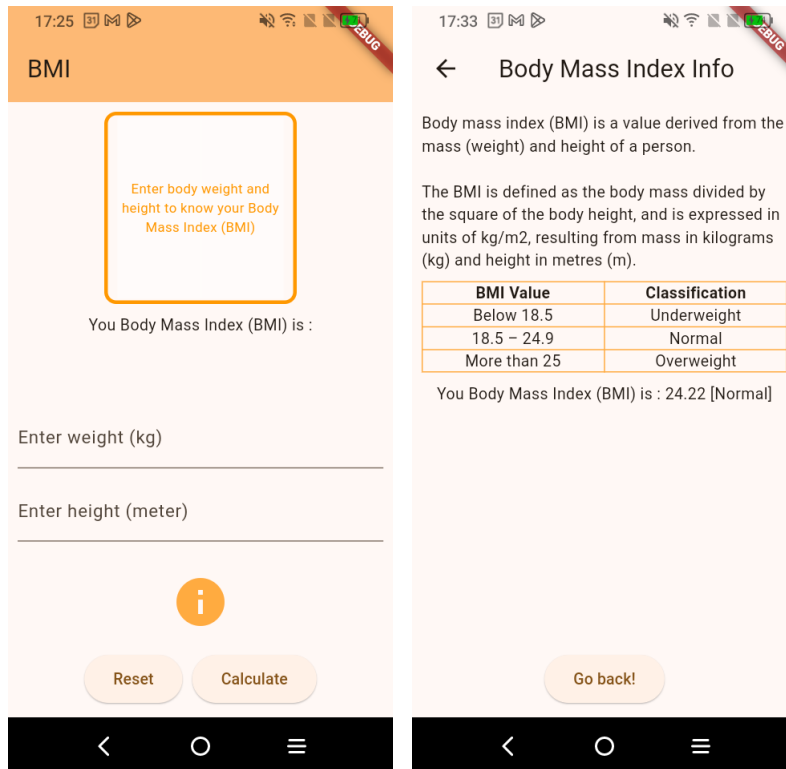


Practical 4: Navigation

We will learn the basic of Flutter navigation in this practical. It allows us to move from one screen to another. Also, we will pass a piece of data from the first screen to the second screen.

We will enhance the BMI calculator with a new Information screen, showing a short description about BMI, the formula to calculate it, and a list of BMI values. Also, we will pass the value of BMI from the main to the information screen.



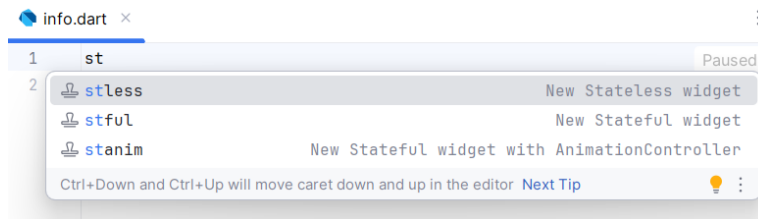
1. Open the BMI project.
2. Modify the **main.dart** file. Insert a new **IconButton** below the **Expanded** widget:

```
//TODO 12: Insert an Expanded widget to take up empty screen spaces
Expanded(child: SizedBox(height: double.infinity)),

IconButton(
  icon: Icon(Icons.info),
  iconSize: 48,
  color: Colors.orangeAccent,
  onPressed: () {
    Navigator.push(
      context,
      MaterialPageRoute(builder: (context) => Info(bmi: _bmiOutput,)));
  },
),
```

3. Right-click on the **lib** folder, select **New -> Dart File**. Assign the name **info.dart** to the new file.

- Within the **info.dart** file, enter **stl** and the system will prompt you a list of widgets, select **stless** to create a new **Stateless** widget. We will not implement any UI update here, therefore, a **Stateless** widget is appropriate.



Assign **Info** as the name of the newly created stateless widget:

```
info.dart x
1  class Info extends StatelessWidget {
2    const Info({super.key});
3
4    @override
5    Widget build(BuildContext context) {
6      return const Placeholder();
7    }
8  }
9
```

- Import the following package:

```
import 'package:flutter/material.dart';
```

- Within the **Info** class, define a local variable named **bmi** to hold the BMI data passing from the first screen. Also, modify the **Info** constructor to include the **bmi** value as a **required** value.

```
class Info extends StatelessWidget {
  //TODO 1: Define a local variable to accept value
  final String bmi;
  const Info({super.key, required this.bmi});
}
```

- Replace the **Placeholder** widget with a **Scaffold** widget. Also, assign a **Text** to the widget showing the heading text.

```
//TODO 2: Replace the Placeholder with a Scaffold
return Scaffold(
  appBar: AppBar(
    title: const Text('Body Mass Index Info'),
  ),
);
```

- Within the **body** of the **Scaffold**, insert a **Padding**, a **Center**, and a **Column** widget.

```
//TODO 3: Insert Padding, Center, and Column
body: Padding(
  padding: const EdgeInsets.all(8.0),
  child: Center(
    child: Column(
      children: [],
    ), //Column
  ), //Center
), //Padding
```

- Inside the **children** of the **Column**, insert the following widgets:

- a. A Text widget to show the general description of BMI.

```
Text('Body mass index (BMI) is a value derived from the mass (weight) and height of
a person.\n\nThe BMI is defined as the body mass divided by the square of the body
height, and is expressed in units of kg/m\u00B2, resulting from mass in kilograms
(kg) and height in metres (m).'),
```

- b. A Table widget to show BMI value and classification.

```
Table(
  border: TableBorder.all(color: Colors.orangeAccent),
  //defaultVerticalAlignment: TableCellVerticalAlignment.middle,
  children: const [
    TableRow(
      children: [
        TableCell(child: Text('BMI Value',
          textAlign: TextAlign.center,
          style: TextStyle(fontWeight: FontWeight.bold)),),
        TableCell(child: Text('Classification',
          textAlign: TextAlign.center,
          style: TextStyle(fontWeight: FontWeight.bold))),
      ],
    ),
    TableRow(
      children: [
        TableCell(child: Center(child: Text('Below 18.5',
          textAlign: TextAlign.center))),
        TableCell(child: Center(child: Text('Underweight',
          textAlign: TextAlign.center))),
      ],
    ),
    //TODO: Continue inserting the remaining two rows.
    ...
  ],
),
```

- c. A Text widget to show the BMI value received from the first screen.

```
bmi==''? Text('Please enter your weight and height.'):
  Text('You Body Mass Index (BMI) is : $bmi'),,
```

- d. An Expanded widget.

```
Expanded(child: SizedBox()),
```

- e. A ElevatedButton widget.

```
ElevatedButton(
  onPressed: () {
    Navigator.pop(context);
  },
  child: const Text('Go back!'),
),
```

10. Save and run the project.



TODO: Enhance the app by including a link to a web URL at <https://data.worldobesity.org/country/malaysia-130/>